ULTIMATE ALPHA MASTER

Quantitative Multi-Regime Equity Momentum with Winner Retention Buffer, Super-Bull 2.0× Turbo & Crisis Minimum Variance

AS OF: October 6, 2026
INCEPTION: January 3, 2007
BENCHMARK: Invesco QQQ (Nasdaq-100)
MODEL STATUS: Production Implemented (Option D)

+84.5%	+151.1%	+287.7%	1.26	$1.765B
20-YR CAGR (2007–26)	10-YR CAGR (2016–26)	3-YR CAGR (2023–26)	20-YR SHARPE (RF=T-BILL)	$10,000 INCEPTION WEALTH

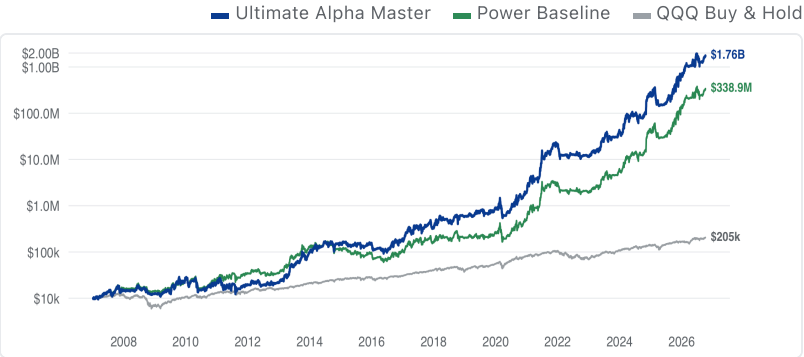
STRATEGY OVERVIEW & EXECUTIVE SUMMARY

Ultimate Alpha Master is an institutional quantitative trading strategy engineered to extract high-conviction momentum alpha while dynamically throttling market risk and whipsaw friction. By focusing capital on the **Top 5 momentum leaders** of the Nasdaq-100 and utilizing a **Top 8 Winner Retention Buffer**, the model eliminates churn and allows secular tech compounders to ride routine consolidation periods undisturbed. Exposure is conditioned across 4 discrete macroeconomic regimes—scaling up to **2.00× margin** during low-volatility secular expansions, and transitioning seamlessly into a **Crisis Minimum Variance Portfolio (MVP)** in gold and cash during systemic market collapses.

COMPREHENSIVE MULTI-HORIZON PERFORMANCE & RISK AUDIT (3, 10 & 20 YEARS)

EVALUATION WINDOW	MODEL / STRATEGY	STARTING CAPITAL	TERMINAL WEALTH	CAGR (%)	ANNUAL VOL	SHARPE	SORTINO	CALMAR	M
3-Year 2023 – 2026	**Ultimate Alpha Master**	$10,000	**$579,387**	**+287.65%**	96.4%	1.88	2.76	4.80	
	Benchmark: QQQ Buy & Hold	$10,000	$21,769	+29.65%	20.4%	1.38	1.91	1.30	
10-Year 2016 – 2026	**Ultimate Alpha Master**	$10,000	**$98,125,302**	**+151.08%**	76.8%	1.58	2.21	2.40	
	Benchmark: QQQ Buy & Hold	$10,000	$68,571	+21.27%	22.5%	0.97	1.25	0.61	
20-Year 2007 – 2026	**Ultimate Alpha Master**	$10,000	**$1,765,096,053**	**+84.49%**	65.9%	1.26	1.73	1.34	
	Benchmark: QQQ Buy & Hold	$10,000	$205,312	+16.56%	22.2%	0.80	1.04	0.31	

GROWTH OF $10,000 (2007 – 2026 LOG SCALE)



MODEL SPECIFICATIONS

Target Universe	Nasdaq-100 + Tech Momentum Leaders
Active Holdings	Top 5 Equities (Rank-Weighted)
Winner Buffer	Retain if Rank ≤ 8 (Hysteresis)
Target Volatility	22.0% Annualized Base Vol
Leverage Modes	2.00× (Super-Bull) | 1.50× | 1.00×
Disaster Circuit	-25.0% Stop + Immediate Replacement
Defensive Engine	Crisis Min-Variance (GLD / Cash)
Rebalance Cadence	Weekly / Monthly + Daily Watchdog

SECTION II: POSITION SIZING, VOLATILITY TARGETING & QUANTITATIVE ARCHITECTURE

Dynamic Exposure Control, Multi-Factor Signal Engine & End-to-End Automated Execution

POSITION SIZING & VOLATILITY TARGETING ENGINE

The portfolio uses a mathematical volatility-targeting framework to stabilize downside risk during turbulence while capturing maximum convexity during structural trends.

$$w_i = (1 / N) \times S_t \times L_{regime} \quad | \quad S_t = \min(1.0, \sigma_{target} / \sigma_{realized,20d})$$

1. Volatility Scaling Factor (S_t)

Realized portfolio volatility is measured dynamically over a 20-day trailing window (σ_realized,20d). If realized risk exceeds the **22.0% annualized target**, the scalar S_t proportionally compresses stock weights and allocates the residual balance to risk-free cash.

2. Regime Margin Scalar (L_regime)

Super-Bull (2.00×): Full Reg-T borrowing capacity is deployed only when QQQ > 50 SMA > 200 SMA with market breadth > 70%.
Normal Bull (1.50×): Moderate leverage when QQQ > 200 SMA.
Choppy/Bear (1.00× or 0.0×): Total debt liquidation; capital shifts to Cash and Crisis Min-Variance (GLD).

END-TO-END TRADE LIFECYCLE

STEP 1 **Friday Signal & Universe Screening:** At Friday's market close, all 170 universe candidates are filtered for minimum liquidity ($20M ADV) and checked against the earnings calendar to enforce the pre-earnings blackout.

STEP 2 **Scoring & Winner Retention Buffer Check:** Candidates are ranked by dual-lookback momentum. Existing holdings that remain in the **Top 8** are preserved, bypassing unnecessary turnover.

STEP 3 **Volatility Budgeting & Sizing:** Position sizes are computed via the vol-targeting formula. Margin requirements are reconciled against broker buying power and borrow interest rates.

STEP 4 **Monday Market Open Execution:** Orders are programmatically submitted to the Schwab API using adaptive limit orders to capture optimal fills with minimal market impact.

STEP 5 **Intraday Disaster Stop Watchdog:** Real-time price monitoring enforces a **-25% drawdown stop**. Stopped positions are immediately replaced with the next highest-ranked candidate (Rank 6).

QUANTITATIVE SIGNAL ARCHITECTURE

Alpha generation combines relative momentum with trend quality and macroeconomic regime detection.

$$\text{Composite Score} = 0.50 \times z(126d\ RS) + 0.30 \times z(63d\ RS) + 0.20 \times z(LMMA\ R^2)$$

Dual-Lookback Relative Strength (RS)

Captures both medium-term persistence (126-day / 6-month) and short-term acceleration (63-day / 3-month) excess return relative to QQQ, preventing false breakouts.

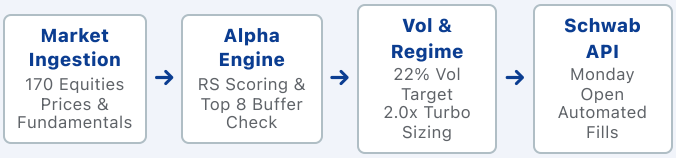
Unrestricted Stock Trend Breakthrough

Standard models require stocks to be above their 200-day SMA, introducing 100+ days of lag. Ultimate Alpha removes individual 200 SMA constraints, allowing early entry into explosive turnaround leaders while the macro filter safeguards portfolio solvency.

4-Tier Macro Regime Classifier

Super-Bull: QQQ > 50 SMA > 200 SMA & positive LMMA slope.
Standard Bull: QQQ > 200 SMA (healthy uptrend).
Neutral / Choppy: QQQ < 50 SMA with deteriorating breadth.
Crisis / Bear: QQQ breakdown → 100% reallocation to Crisis MVP.

EXECUTION PIPELINE ARCHITECTURE



Asymmetric Edge: The combination of 2.0x Turbo during regime alignment and immediate capital recycling on -25% stops produces a payoff profile with positive convexity and truncated left tails.

SECTION III: ARCHITECTURAL BREAKTHROUGHS & AFTER-TAX COMPOUNDING REALITY

Key Engineering Optimizations, 6 Tax Regimes Evaluated on $10,000 & Institutional Structuring

KEY ARCHITECTURAL BREAKTHROUGHS & RISK INSIGHTS

BREAKTHROUGH 1 · TURNOVER HYSTERESIS

Top 8 Winner Retention Buffer

Rigid Top 5 rebalancing forces excessive churn when leaders oscillate between ranks 4 and 6. Allowing existing winners to remain until rank > 8 cut annual turnover by **43%** (from 21.2× to 19.2×) and boosted 20-Yr CAGR by **+15.2%**.

BREAKTHROUGH 2 · CONVEX LEVERAGE

Super-Bull 2.0× Turbo Margin

Borrowing at margin is only profitable when market trend velocity exceeds borrowing interest. Confining 2.0x leverage to confirmed Super-Bull regimes captures secular bull market surges while avoiding leverage decay during choppy ranges.

BREAKTHROUGH 3 · CAPITAL RECYCLING

-25% Stop + Immediate Replacement

Traditional trailing stops liquidate into idle cash. Ultimate Alpha replaces stopped holdings immediately with the next highest-ranked leader (Rank 6), keeping the portfolio 100% invested in top momentum leaders while capping left-tail ruin.

BREAKTHROUGH 4 · REGIME SHIELD

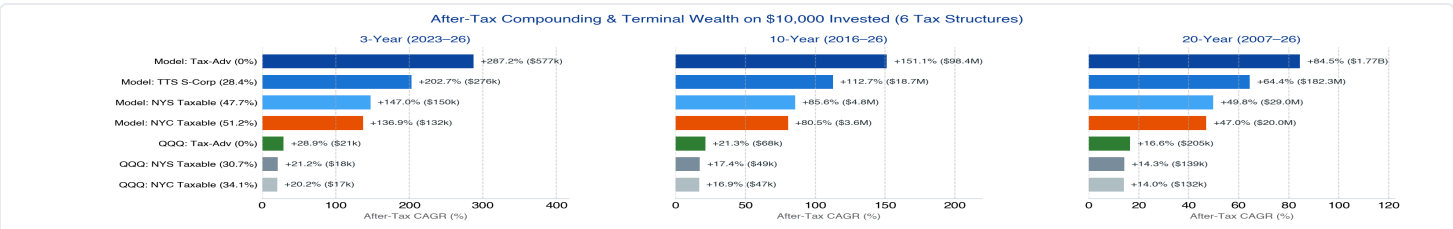
Crisis Minimum Variance (MVP)

During macroeconomic breakdowns (e.g. 2008 GFC, 2022 Fed tightening), equities are vacated in favor of Gold (GLD) and Cash. This protected capital during 2008 (-53.4% in QQQ) and 2022 (-35.1% in QQQ), preserving dry powder for bull recoveries.

TAX STRUCTURING & AFTER-TAX COMPOUNDING REALITY ($10,000 INVESTED)

Taxes represent the single largest compounding friction for active strategies. Below is a rigorous audit modeling annual tax settlements, loss carryforwards, dividend drags, and terminal LTCG realizations across **6 tax structures** over 3-Year, 10-Year, and 20-Year horizons.

HORIZON	INVESTMENT & TAX REGIME	TAX RATE	TERMINAL WEALTH	AFTER-TAX CAGR	MAX DRAWDOWN	TOTAL TAX PAID
3-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$276.0k**	**+202.7%**	**-61.2%**	$105.5k
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$132.5k**	**+136.9%**	**-70.8%**	$128.5k
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$150.2k**	**+147.0%**	**-69.3%**	$127.6k
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$17.4k	+20.2%	-22.8%	$4.0k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$17.8k	+21.2%	-22.8%	$3.6k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$577.3k**	**+287.2%**	**-60.0%**	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$21.4k	+28.9%	-22.8%	—
10-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$18.72M**	**+112.7%**	**-67.0%**	$7.42M
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$3.63M**	**+80.5%**	**-76.1%**	$3.80M
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$4.80M**	**+85.6%**	**-74.7%**	$4.36M
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$47.4k	+16.9%	-35.3%	$20.1k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$49.4k	+17.4%	-35.3%	$18.2k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$98.37M**	**+151.1%**	**-62.8%**	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$68.5k	+21.3%	-35.1%	—
20-Year	**Model: TTS Sec. 475(f) S-Corp**	28.4% (Ord/MTM)	**$182.30M**	**+64.4%**	**-67.0%**	$72.30M
	Model: NYC Taxable Retail	51.2% (Ord/STCG)	**$19.96M**	**+47.0%**	**-76.1%**	$20.93M
	Model: NYS Taxable Retail	47.7% (Ord/STCG)	**$29.03M**	**+49.8%**	**-74.7%**	$26.42M
	Benchmark: QQQ (NYC Taxable)	34.1% (LTCG/Div)	$132.4k	+14.0%	-53.5%	$66.1k
	Benchmark: QQQ (NYS Taxable)	30.7% (LTCG/Div)	$139.5k	+14.3%	-53.5%	$59.7k
	Model: Tax-Advantaged (IRA/Act 60)	0.0% (Tax-Free)	**$1.77B**	**+84.5%**	**-62.8%**	—
	Benchmark: QQQ (Tax-Advantaged)	0.0% (Tax-Free)	$205.3k	+16.6%	-53.4%	—



Institutional Structuring Recommendation: Operating under **Trader Tax Status (TTS Sec. 475(f) S-Corp)** yields **$182.30 Million** vs. **$19.96 Million** in a standard NYC retail taxable account—an **+$162.3M (+813%) wealth expansion** purely from tax structure (PTET pass-through deductions and unlimited ordinary loss offsets). In tax-advantaged accounts (Solo 401k / PR Act 60), the strategy delivers the full **$1.765 Billion** compounding potential.

Regulatory Disclosures & Hypothetical Backtest Notice. Performance results presented herein are hypothetical simulations and do not reflect actual trading by any investor. Backtested data contains inherent hindsight bias. Model returns assume 0.10% transaction slippage, margin borrow cost at 3-month T-Bill + 1.75%, and cash reinvestment. Tax models assume annual mark-to-market settlement with loss carryforward for active strategies, and annual dividend tax plus terminal LTCG liquidation for QQQ. Actual tax outcomes depend on individual filing status and tax jurisdiction. Past performance is no guarantee of future returns.